GOVERNMENT ENGINEERING COLLEGE, RAJKOT

SUBJECT: DMGT(BE04000261) *SEMESTER -4* *BRANCHES: C.E., A.I.&D.S.*

TUTORIAL NO 5

Topics: (1) *Basic of Graphs (2) to Matrices (and vice versa), Warshall's algorithm & (3)* Various types of Trees, MST using Kruskal's & Prim's *algorithm from Unit 5 of the syllabus for the A.Y. 2025-26*

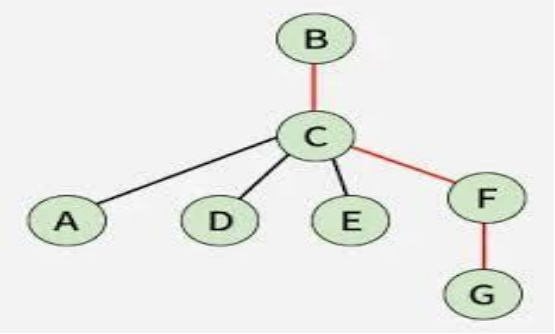
Qn-1: Define Path, Cycle, Simple graph, Directed graph with two examples for each.

Qn-2: Define Regular, Complete, Bipartite and Connected graph with two examples for each.

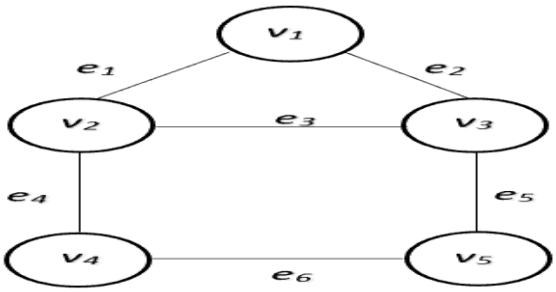
Qn-3: Draw a graph with 7 vertices each of degree odd, if possible. In case of no possibility justify your reason.

Qn-4: Define Isomorphic graphs. Give(Draw) an example of two graphs(with vertices 5) which are isomorphic to each other. Give(Draw) one more example of two graphs(with vertices 4) which are not isomorphic to each other with proper justification.

Qn-5: Define Geodesic and find all geodesics between the vertices A and G from the following graph.



Qn-6: Find the (i) Adjacency matrix and (ii) Incidence matrix for the following graph.



Qn-7: Draw the corresponding graph from the given Adjacency matrices.

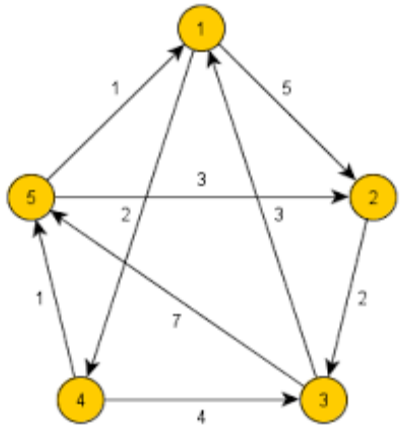
(i) [0 1 1 1 0; 1 0 0 0 1; 1 0 0 0 1; 1 0 0 0 1; 0 1 1 1 0]

(ii) [0 1 0 1; 1 0 1 1; 0 1 0 1; 1 1 1 0]

Qn-8: Define Weakly Connected, Strongly Connected and Unilaterally Connected graphs with an example for each.

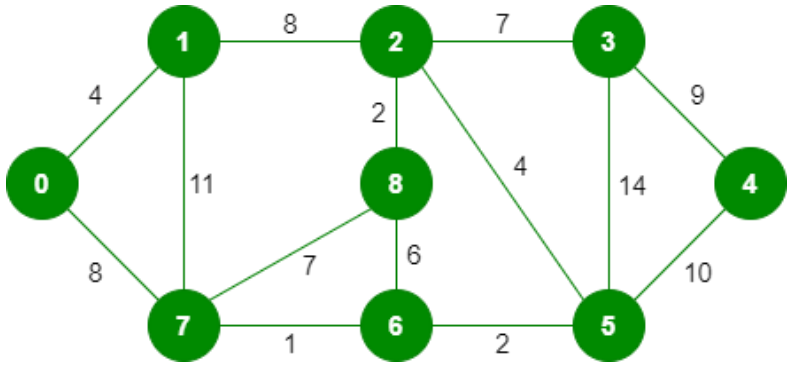
Qn-9: Define strong and week components of a graph with an example.

Qn-10: Use Warshall's algorithm to find the shortest distance between each pair of two vertices for the given graph.



Qn-11: Define Tree, Binary tree, Rooted tree, Spanning tree with an example.

Qn-12: Use Kruskal's algorithm to find the MST(Minimal Spanning Tree) for the given graph G.



Qn-13: Use Prim's algorithm to find the MST(Minimal Spanning Tree) for the given graph G, mentioned in above question.

Qn-14: Find all the spanning trees for a complete graph K_4.

Qn-15: Draw a tree T with atleast 10 vertices and atleast 2 level, then write its pre order and post order traversal.